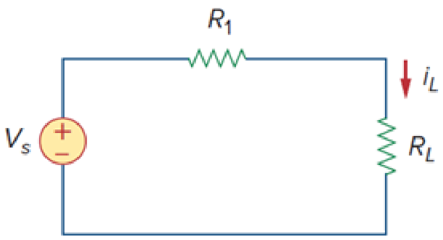
# Chapter 2, Problem 71P

## Problem

Figure 2.131 represents a model of a solar photovoltaic panel. Given that *Vs* = 95 V, *R*1 = 25 Ω, and *iL* = 2 A, find *RL*.

**Figure 2.131:**




## Step-by-step solution

**Step 1** of 2

Consider the circuit of textbook Figure 2.131.

It is given that $V_s = 95\ \text{V}$, $R_1 = 25\ \Omega$ and $i_L = 2\ \text{A}$.

Apply Kirchhoff's voltage law to the circuit.

$$-V_s + i_L R_1 + i_L R_L = 0$$
$$V_s = i_L R_1 + i_L R_L$$

**Step 2** of 2

Substitute $95\ \text{V}$ for $V_s$, $25\ \Omega$ for $R_1$ and $2\ \text{A}$ for $i_L$.

$$95 = (2)(25) + 2R_L$$
$$2R_L = 95 - 50$$
$$2R_L = 45$$
$$R_L = 22.5\ \Omega$$

Therefore, the value of unknown resistance $R_L$ is $\boxed{22.5\ \Omega}$.